IGBT Sentinel | Pre-installation Test Cell

NU–WABTEC Hackathon 2026 • Engineering concept and demonstration package

STATUS: PROPOSED DESIGN. The supplied dashboard uses synthetic data only; no physical IGBT was tested.

1. Problem and outcome

A locomotive IGBT may be found defective only after installation, causing avoidable labor and downtime. Our design screens a module beforehand and produces traceable results. This package demonstrates the operator workflow, not an electrically validated tester.

2. Requirements traceability

Physical tester concept; multiple parameters and pass/fail decision; laptop report; selected parameters and time justification; technician procedure; approximate cost. All six are addressed as a design proposal; the software demo addresses the interface/reporting workflow.

3. Concept architecture

Module → enclosed and keyed test fixture → qualified programmable instruments → independent safety/interlock controller → isolated data acquisition → laptop dashboard and report. The safety controller shall be independent of the reporting application. No high-voltage output is enabled without a validated fixture, door interlock, emergency stop and discharge verification.

4. Tiered verification

Quick profile: proposed 3 min target, screening gate threshold, saturation voltage, collector leakage, gate leakage and diode forward voltage. Advanced profile: proposed 12 min target, adding gate driver rails, case temperature and (in a production design) approved switching / insulation tests. Targets are engineering assumptions pending timing trials, not experimentally confirmed performance.

5. Important engineering restriction

The quick / advanced test plan does not reproduce operating conditions in full. Production dynamic and insulation tests require specialized instruments, validated connections,

current-limited fixtures, safe discharge processes, qualified personnel, and device-specific limits. An unqualified PASS is not a release for locomotive installation.

6. Parameter prioritization

Parameter	Reason	Category
VGE(th)	Detect altered gate threshold	Static
VCE(sat)	Screen abnormal on-state voltage	Static under rated conditions
ICES	Identify blocking/leakage issues	Static / voltage-limited
IGES	Screen damaged gate insulation	Static
VF	Screen free-wheeling diode	Static
Gate drive ± rails	Check gate-drive subsystem interface	Integration
Desaturation response	Protection behavior, specialized validation	Dynamic, professional lab
VISOL	Isolation between terminals/baseplate	HV specialist test
Thermal behavior	Assess temperature-dependent performance	Extended lab test

Illustrative dashboard thresholds are invented solely for simulated demonstration and must NOT be used for any real unit. Obtain the exact Wabtec module part number/datasheet and an approved test specification before setting pass/fail limits.

7. Technician workflow (proposed)

1. Confirm module ID, revision and approved test recipe.
2. Complete lockout/tagout, visual inspection, grounding and interlock checks under site procedure.
3. Connect module using a certified keyed fixture; do not improvise wiring.
4. Run approved static test recipe using electrically rated, calibrated equipment.
5. If permitted by safety review, run advanced instrument-controlled tests.
6. Review out-of-limit or missing measurements; quarantine failed or uncertain modules.
7. Archive signed report and release only under approved engineering acceptance rules.

8. Indicative budget (USD, non-quoted allowances)

Component	Low $	High $
Enclosure and interlocks	600	1400
Keyed fixture, connectors, insulation	600	1600

Low-voltage instrument / DAQ	350	1100
Current-limited bench supply	250	800
Controller / interfaces / E-stop	220	650
Calibration, assembly, spares	700	1800
Laptop / PC (if required)	0	900

Low-voltage conceptual bench: $2,720–$8,250. This EXCLUDES an industrial high-power switching bench, certified HV insulation tester, professional safety certification and taxes/shipping. No supplier quotations were obtained. Obtain vendor quotes before procurement.

9. Time and economic value

Example ONLY: if replacing one installed failed module would consume 6 technician-hours versus 0.5 technician-hours for a bench screen, the gross difference is 5.5 technician-hours per caught defect. Multiply by documented actual labor rate and real defect incidence. This is not a measured Wabtec saving. Time targets require stopwatch-based validation across good and faulty samples.

10. Demonstration script

Open IGBT_Sentinel_Demo.html in Chrome/Edge. Run Healthy → PASS; Collector leakage → FAIL (ICES); Gate failure → FAIL (VGE(th) and IGES). Export CSV or Print → Save as PDF. Explain the labels “SIMULATED DATA” and how actual instruments would eventually replace the synthetic feed.

11. Further validation / acceptance gates

- Request exact module electrical part number and authorized pinout from Wabtec.
- Agree datasheet-based limits at specified test current, voltage and temperature.
- Perform failure-mode and effects analysis, insulation/clearance review and safe-state testing.
- Compare readings with traceable calibrated reference instruments.
- Record false-pass / false-fail performance on blind samples.
- Obtain qualified electrical engineering approval before energizing power devices.

References

Wabtec, NU–WABTEC Hackathon 2026 case briefing, supplied slide deck (especially slides 5, 8–11).

Infineon Technologies, “Outgoing Quality – 100% Final Test of IGBT Modules” (2020), https://www.infineon.com/assets/row/public/documents/faq/2020-03-03-final-test-igbt.pdf

Infineon Technologies, “Industrial IGBT Modules: Explanation of Technical Information,” AN 2011-05.